

Geographic Roots & Academic Studies: How Environments Shape our College Choices

Introduction

The purpose of this study is to investigate and compare BYU student’s geographic backgrounds and their choice of academic colleges. Comparing home environment factors and distance from BYU examines an important concept within higher education geography. College choices made by students affect institutional planning, workforce development, and resource allocation, creating a valuable geographic problem with implications for BYU’s admissions and student services. Prior research laid the foundation that various factors work together to shape academic trajectories. Social circles and familial guidance have a profound impact on choices of major and often outweigh financial considerations (Aldawsari, 2024). Proximity to educational institutions plays a role in both academic performance and course selection (Brownie et al., 2023). Another study showed that geographic mobility varies between female and male students with female students being less likely to relocate for education (Farré, Ortega, 2024). A study done at BYU found that a sense of belonging also influences BYU student’s major selections (Hunsaker, 2024). This study seeks to examine how home environments and distance from BYU campus may impact academic paths at BYU.

Methods

Our student group distributed questionnaires to other students using Google Forms. Each member of the group sent out the questionnaire consisting of a set of demographic questions then the question sets from each group member. The questionnaire questions for my data:

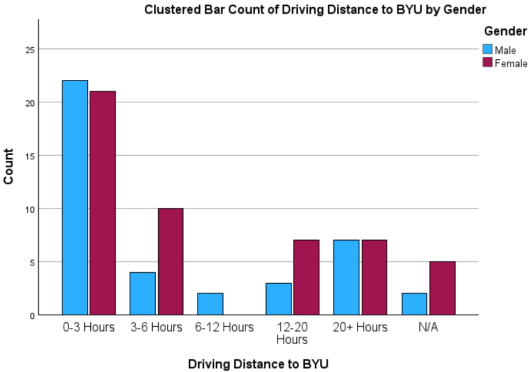
1. What type of environment did you grow up in?
Suburban, Urban, Rural, Other
2. How many hours would it take you to drive from your home to BYU campus?
0-3, 3-6, 6-12, 12-20, 20+, N/A
3. Which BYU college is your major in?
College of Family, Home, and Social Sciences, Ira A. Fulton College of Engineering, College of Life Sciences, Marriott School of Business, College of Humanities, College of Physical and Mathematical Sciences, College of Fine Arts and Communications, College of Nursing, College of Education, College of Religious Education, Undecided/Undeclared

This method of distributing questionnaires could introduce bias as each student is likely to send the questionnaire to people they know, and these people may have similarities. This could be corrected in the future by distributing the survey to a more diverse sample of people unaffiliated with the student. In addition to this, the questionnaire data only yielded (n=96) responses which is inherently a small sample that limits statistical power. This could be avoided moving forward by gathering a larger sample size. In addition to this, self-reporting of distances may contain measurement errors, this can be avoided by using zip codes or more standard forms of measurement.

Mann-Whitney U Test

Does gender impact distance from BYU campus?

- H₀: There is no significant difference in distance from BYU between male and female students.
- H_a: There is a significant difference in distance from BYU between male and female students.

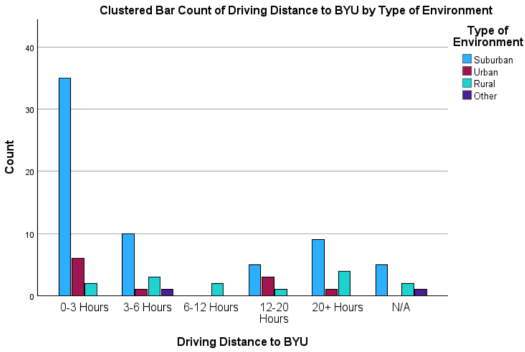


A Mann-Whitney U (MWU) test was conducted to compare distance from BYU campus between both male and female students. The MWU test gave a U statistic of 886.00 and a significance level of $p = .324$. This p-value exceeds the alpha level of .05, indicating the results do not meet the 95% confidence threshold for statistical significance. The null hypothesis is accepted which means that there is no significant difference in distance from BYU between genders ($U = 886.00$, $p = .324$). Male and female students in this sample live at similar distances from campus. These findings contrast with the observations of previous studies regarding gender differences and geographic mobility

(Farré, Ortega, 2024), possibly due to BYU’s admissions and recruitment strategies with BYU recruiting heavily in specific areas such as the “Pioneer Corridor”. Future research may benefit from examining this phenomena across other institutions.

Kruskal-Wallis H Test

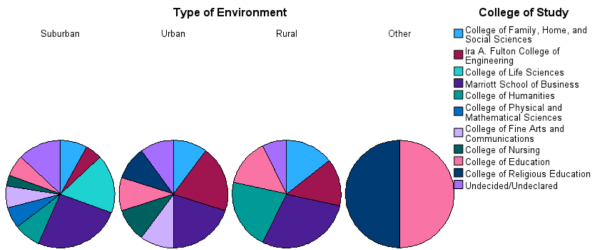
- Does distance from BYU differ across students from different environment types (urban/suburban/rural)?
- H₀: There is no significant difference in distance from BYU across different environment types.
- H_a: There is a significant difference in distance from BYU among students from different environment types.



The Kruskal-Wallis H (KWH) test was conducted to examine whether distance from BYU differed across various environment types (urban, suburban, rural). The analysis revealed a test statistic of $H = 8.79$ ($df = 3$) and a significance level of $p = .032$. The p-value found is below the required alpha level of .05, meeting the 95% confidence threshold of statistical significance. The null hypothesis is rejected meaning that there is a significant difference in distance from BYU across home environment types ($H = 8.79$, $p = .032$). Students from different home environment types (urban, suburban, rural) differ significantly in their driving distances to BYU campus. This finding aligns with prior research that geographic location affects educational choices and access (Brownie et al., 2023). This may also be due to BYU’s recruitment strategies with heavy recruitment in specific areas and overall interest being higher in certain states. Future research would benefit from additional variables like cost of attendance or transportation access to more carefully examine these geographic phenomena.

Chi-Square Test

- Does home environment type impact BYU student’s college of study?
- H₀: There is no significant difference in the distribution of students across colleges among different home environment types.
- H_a: There is a significant difference in the distribution of students across colleges among different home environment types.



A chi-square test of independence was conducted to compare the distribution of students across colleges among home environment types at BYU. The Pearson chi-square statistic was $\chi^2 = 47.31$ ($df = 30$) with a significance level of $p = .023$. This p-value is below the alpha level of .05, suggesting a statistically significant difference in distributions at the 95% confidence level. While the p-value suggests rejecting the null hypothesis ($\chi^2 = 47.31$, $p = .023$), because 86.4% of cells had expected frequencies < 5 , we cannot determine if there is a significant difference in the distribution. While the p-value may suggest there is a significant difference in the distribution, which would support geographic theories about how place shapes identity and interests, we cannot draw any conclusions without meeting the required test assumptions. Future research should collect a larger sample or combine categories to help meet the number of expected frequencies for each cell, ensuring statistical validity. Further analysis could explore why certain environment types, such as ‘rural,’ show different distribution patterns across majors compared to other environment types. In addition, understanding what the category of ‘other’ means in terms of home environment type could also be useful knowledge.

References

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Brownie, S., Yan, A.-R., Broman, P., Comer, L., & Blanchard, D. (2023). Geographic location of students and course choice, completion, and achievement in higher education: A scoping review. *SAGE Open*, 13(3).

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